Adaptive Routing for Vector and Graph-Based Retrieval-Augmented Generation

by

William Barrett

This thesis has been submitted in partial fulfillment for the degree of Master of Science in Artificial Intelligence

in the
Faculty of Engineering and Science
Department of Computer Science

March 2026

Declaration of Authorship

This report, Adaptive Routing for Vector and Graph-Based Retrieval-Augmented Generation, is submitted in partial fulfillment of the requirements of Master of Science in Artificial Intelligence at Munster Technological University Cork. I, William Barrett, declare that this thesis titled, Adaptive Routing for Vector and Graph-Based Retrieval-Augmented Generationand the work represents substantially the result of my own work except where explicitly indicated in the text. This report may be freely copied and distributed provided the source is explicitly acknowledged. I confirm that:

- This work was done wholly or mainly while in candidature Master of Science in Artificial Intelligence at Munster Technological University Cork.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at Munster Technological University Cork or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this project report is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed: _____

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Abstract

Faculty of Engineering and Science
Department of Computer Science

Master of Science

by William Barrett

Retrieval-Augmented Generation is how we dynamically integrate external knowledge in to our agentic systems. LLMs are trained on large corpuses of data but hallucinate when met with information they are not trained. RAG performs well in specific chunk based lookups. When external knowlege bases are large howerver, RAG fails to at looking at thematic data across the whole corpus. Microsoft introduced graphrag as a solution to this issue. GraphRag prebuilds a graph model outline entities themes and relationships acorss the entire corpus. Modern research focuses on the integration of RAG or GRAPH rag. A current gap is that there is not much research looking at how to integrate both simulaatniously. Current research takes a binary approach of this or that. in some case they eveny split results 50/50. The purpose of this research is to build and emperically evaluate routing strategies to address this gap. these straties will attempt to identify incoming queries and determine an adaptive routing strategy that determines the best means of blending the results of both vector and graph based RAG. The contribution of this research will include an analysis of multiple classification strategies, and investigation into blending strategies and and investigation into if adaptive blending out performs indivual retrievaal strategies.

Acknowledgements

The acknowledgements and the people to thank go here, don't forget to include your project supervisor...

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Abbreviations

LAH **L**ist **A**bbreviations **H**ere

For/Dedicated to/To my...

Chapter 1

Introduction

1.1 Research Context

1.1.1 Background

1.1.2 Teory

1.1.2.1 Related Work

1.2 Research Aim

blah bna;l [1]

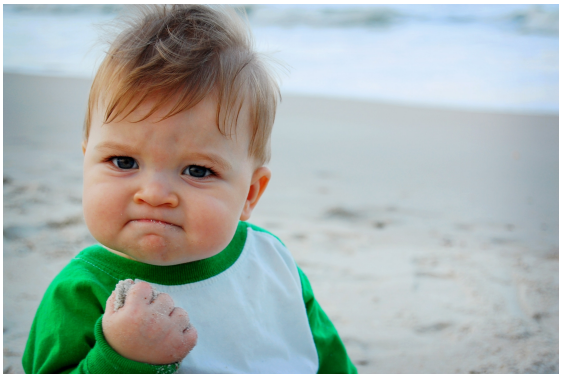


FIGURE 1.1: My first figure

Objective	Description
Objective 1	Description of objective 1
Objective 2	Description of objective 2
Objective 3	Description of objective 3

Table 1.1: Research Objectives

1.3 Research Objects

This chapter should provide background description for your project. This chapter should comprise around 1000 words. Here you are setting the scene, remember the reader may know little about your project at this stage (other than the abstract). N.B. The chapter sections outlined in this document are suggested, some projects will have a greater or lesser emphasis on different sections or may change titles and some will have to add other sections to provide context or detail.

1.4 Motivation

Provide motivation for reading the report. Why is it important to do a project on this topic? This should cover your key motivation for this. This section can include the personal pronoun but **the rest of the report should be third person passive**, this is the case with most technical reports! For example here it is fine to say "... I decided to develop and app to help ...".

1.5 Contribution

Specify the problem clearly. List your project goals and research questions here. Enumerate the main contributions. How have you advanced the state of the art i.e what have you done that is new?

1.6 Structure of This Document

Introduce the structure of your report. This section is quite formulaic. Briefly describe the structure of this document, enumerating what does each chapter and section stands for. For instance in this work in Chapter 2 the guidance in structuring the literature review is given. Chapter 3 describes the main requirements for the problem definition and so on ...

Chapter 2

Literature Review

The key question to answer in this chapter is: "What has been done/is being done".

You must present a summary of background research - what has been done in this area; any related work. Provide reader with information to understand the rest of the report. This chapter comprises around 3000 words and should put your project into context within the field. You must identify the main sources of information you should be aware of within your chosen area and use this activity to strengthen your knowledge in this area. So here you should develop a knowledge of not only your specific topic but also how it is changing.

Include relevant references. For the (rare) projects with no implementation component, e.g., using neural networks to explore or extend the properties of some mathematical model, or to make predictions of some kind, one might break this chapter into two and write a chapter on neural networks and another on the area of application.

To conduct this activity you should aim to read most of a relevant paper every other day for the remainder of the project, make a brief summary of the main findings and how they relate to your work (less than 30 words). You should continue this activity after your project as CPD.

You should begin your journey of discovery in reverse order to the listing above (which is given in order of academic importance/significance). So when you are researching your topic first look up some TedX talks or youtube tutorials, then research what companies are doing in the area, then get a handful of very good texts on the core topics of your area (anything older than 5 years usually is not helpful here) and finally start reading conference or journal papers (again newer is better here). In particular during this section you may need to use tables to list resources. These are also automatically formatted in latex thus allowing you to concentrate on content. for example table 2.1.

Parameter	PET
Youngs Modulus	2800-3100MPa
Tensile Strength	55-75MPa
Glass Temperature	75°C
Density	1400kg/m^3
Thermal Conductivity	0.15-0.24Wm^{-1}K^{-1}
Linear Expansion Coefficient	7×10^-5
Relative Dielectric Constant @ 1MHz	3
Dielectric Breakdown Strength	17kVmm^{-1}

Table 2.1: PET Physical Properties

Make sure you cite[2] as many **relevant** papers [3], luckily Latex will take care of the ordering of the citations [4] for you.

The aim here is that you find the trends in your topic (3), and more in general in the area in which your topic resides (2) your project falls under and from these trends you develop your initial project question further and begin to get insights into how others have solved/approached similar problems. Think of this section as colouring in your initial idea. Before you approach this section you should read at least 4/5 good review papers. Also remember if you find that you are detailing topics that you have not introduced earlier in this section you need to add a reference for background reading.

Chapter 3

Design

The key question to be addressed in this chapter is: "How do I plan to achieve what I have outlined in the previous chapter".

This chapter should include requirements capture, data flow, entity relation, object diagrams, etc.; overall system structure, comparison and justification of choices for algorithms and data structures. Critical evaluation is to be done in a later chapter. Again, all sections below are suggestions and will vary significantly from project to project, the key element to be addressed is the core question of the chapter.

3.1 Problem Definition

Describe the problem you are trying to solve in this project in technical detail, here you need to be expicit as details will impact the design. Sometimes here there will be a need at some point during the report to display an equation that may be core to your project. For example if the project is on gait detection what equation are you using to determine gait? If the project is on localization what is the method/formula? The formatting of these is reliably done in Latex also as we can see in equation 3.1.

$$\frac{d}{dt}(\frac{\partial L}{\partial \dot{c_i}}) - \frac{\partial L}{\partial c_i} + \frac{\partial P}{\partial \dot{c_i}} = F_i, \tag{3.1}$$

3.2 Objectives

Enumerate the objectives you want to achieve in your project. Again as this is an early stage these will tend to change but there should be a rational explanation for this

change. Always document your work, keep a lab book during the term that you only use for FYP!

3.3 Requirements

Enumerate the functional requirements you want your project to have.

Please, do not include use cases here! If you want to create a one-to-one mapping between functional requirements and use cases (which does not necessarily need to be the case, indeed most likely this will not be the case) do it elsewhere. Here should purely describe **what do you want to do**. In no case should you use this section to provide a description of how to implement them, that is for later. For people doing projects that are not heavy implementation projects (e.g. deploying an architecture or testing a novel tool in specific conditions) this structure can still be used as it will force you to think about what you plan to achieve and what possible metrics you may need to measure success.

Let me explain this with more detail: A common mistake is that people confuse the problem description with the solution approach. This is a common mistake by confusing the *what* with the *how*. Here we are purely focused on the what: What is this project about? What are the objectives? What are the functional and non-functional requirements?

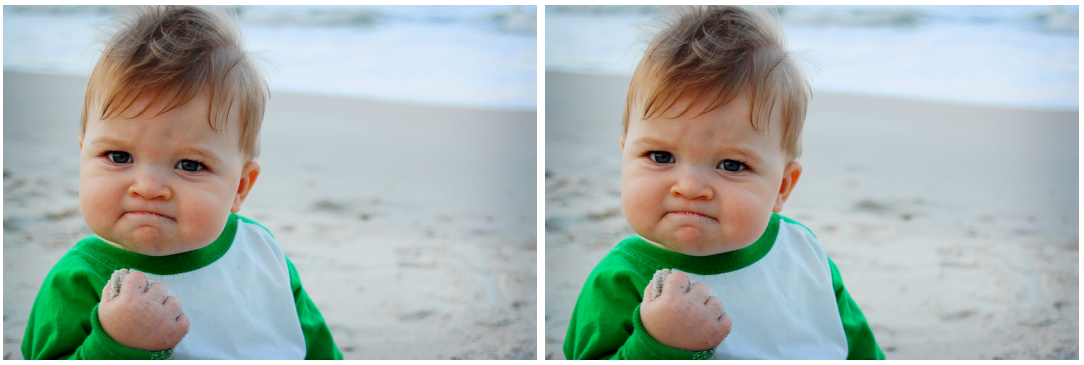
How are we going to do all these things? Well, this is a question for next section. Provided a problem, an objective or a functional requirement, obviously there will usually be many ways of doing it, thus there will be many *hows*, but the definition, the *what* we want to achieve will be unique. LAteX hint: One other display structure you may wish to use at some stage during the report is a figure array. This can also be easily done with Latex and is shown in figure 3.1

Make sure to also enumerate the non-functional requirements you want to achieve in your project (i.e. broadly speaking how your system will operate).

3.4 Design

Describe the architecture of the solution that you have in mind, including technologies involved (e.g., frameworks, programming language).

Provide a high level view of the system and then break down of the system. Include any packages or components you are using (the versions of these should be given) and what



(a) Figure A (b) Figure B

FIGURE 3.1: Two Success kids

they are responsible for and what they communicate to. You need to be as specific here as you can. If you have hardware element in your project this is also where you provide a high level view of how these elements integrate into the project. So for a project that is cyber-physical you will have both a hardware and software architectural diagram.

Chapter 4

Implementation

This chapter should include the most relevant/important code listing and discuss relevant/interesting aspects; explain features you wish to highlight.

4.1 Important Features

For each aspect you could provide the following info:

1. Description of the difficulty: Brief description of the problem you found.
2. How did it affect the original project design?: Indicate how this difficulty affected:
   (a) the architecture of your solution
   (b) if it represented a risk to your overall project (i.e. was it a FR)
   (c) if it affected your methodology to develop your project (i.e. did developing this aspect change the design)
   (d) if it changed your implementation schedule (i.e. did it delay other items)
   (e) if it changed the evaluation plan (i.e. was more/less testing now required?)
   (f) If there is complex code/configuration details discuss it here also.

From January to September you will be developing your project. Along the way you will have encountered difficulties which might have modified your original plan so that you can come up with an actual developed project. You can discuss some of these that led to significant changes (if they exist). It is important here is that there is a clear rationale for design and implementation decisions made.

Chapter 5

Testing and Evaluation

The goal of this chapter is an objective evaluation of the final system. The evaluation must be quantitative and not qualitative. Indicate how the system was verified/tested. You may perform qualitative evaluation but this should not form the basis of the main conclusions you derive from the evaluation. This evaluation, where possible, should be comparative, i.e. you should evaluate your system against a commercially available system and/or system detailed in a research publication. If no implementation was done, you could describe the types of experiments/simulations that were carried out. Explain why these and not others were performed and discuss the most important parameters in the tests/simulations and describe their roles.

You should demonstrate operational testing of the project using real or contrived data sets to evaluate aspects of the project not encompassed in the software testing (e.g. quantify how well does your project achieved the overall goal).

- For software based projects this will include, but should not be limited to, evaluation of non-functional requirements.
- For infrastructural projects this testing should include system/network KPI analysis.
- For analysis based projects (ML, malware or other) this may include model evaluation or YARA rule validation, for example.
- For management projects, where software testing or infrastructure testing may not be in scope, the test process for the system is expected to be more rigorous and well described than a project incorporating significant development work.

Summarize the output data, and the statistical or other techniques to deduce your results. Summarize your results, including tables or graphs as appropriate with a brief

description of each. Here, where possible, compare your results with other products/systems. Identify any possible threats to the validity of your results, and discuss each briefly here (you will discuss in more detail in the next chapter).

Chapter 6

Discussion and Conclusions

Write a critical evaluation of your work. You should state your goals and research questions again and outline in short how they were or were not met or answered. What have you achieved? What more could have been accomplished? Mention any problems that affected the progress of your work. Is there any scope for extension/improvement/scalability? What are the new challenges to be faced in the future? The expectation here is that you should discuss the results presented in the previous evaluation section of the project in their totality (i.e. as a whole) from which you will then draw clear conclusions both on the quantitative and qualitative aspects of the overall project. The conclusion section of this chapter should conclude the project.

6.1 Discussion

Discuss how well your solution solves the problem, based on your results from the evaluation chapter. Discuss how well you addressed the project, and what you might do differently if you were to do it again. Make sure to identify how you handled any problems that arose during the project.

6.2 Conclusion

Enumerate the main conclusions you have got in terms of background, problem description and the solution approach you have come up with. Detail your primary and any secondary conclusions from your project.

6.3 Future Work

Discuss any proposals for completion of the project, or for enhancements, or for re-design of your solution or software. Enumerate all the things you would have wanted to do should you have more time to work on this project.

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Appendix A

Code Snippets

Put appendix material in this section e.g. code snippets

If there is too much code to include in the Appendix, submit the rest on a USB disk or CD. Some code should be easily available for examiners to assess.

USE THE APPENDICES

Appendix B

Wireframe Models